Bart Olsthoorn

Ph.D. in Physics · Product Engineer at Flower · Founder of Glimt

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Profile

I'm a full-stack software engineer with a physics Ph.D. I build product end-to-end, from data pipelines on Databricks to backend APIs and the web interfaces on top, and do much of that work with AI agents as a development tool. I'm equally comfortable leading a small team or training a model.

Experience

2026 – now
Glimt
Founder · Sweden
Building scientific instruments, starting with a scanning tunnelling microscope for atomic-resolution imaging.

2023 – now
Flower
Stockholm · Python, Go, AWS, Databricks

Feb 2026 – now
Product Owner / Product Engineer · Distributed Assets
Owning roadmap and prioritisation for the team; building our systems with AI coding agents.

Dec 2024 – Feb 2026
Software Engineer · Distributed Assets
Built Flower Bridge, the integration platform for distributed energy assets (residential and C&I batteries, solar, EVs, heat pumps), end-to-end: Databricks data pipelines, backend APIs, and web interfaces.

Jan 2024 – Dec 2024
Engineering Manager (interim) · Asset Management
Technical recruiting and team leadership during a growth phase.

Feb 2023 – Jan 2024
Software Engineer · Asset Management
Built a platform supporting the Nordic electrical grid with grid-scale solar and battery systems.

2012 – 2017
Freelance Software Engineer
Stockholm, Sweden and Leiden, The Netherlands
Shipped production software across a range of stacks and domains: early generative AI with PyTorch / GANs at *Similar.ai*; Ruby and Elixir work on search, recommendations, and deduplication at *Universal Avenue* (now Velory); scalable Rails SaaS at *InnerBalloons* (acquired by Yext) and web-data systems at *Pointer Brand Protection*.

Education

2018 – 2023
Ph.D., Physics · KTH Royal Institute of Technology
Thesis: *Homology and machine learning for materials informatics.* 11 peer-reviewed publications.

2015 – 2018
M.Sc., Computational Physics · Stockholm University

Selected publications

2023
Persistent homology of quantum entanglement
Physical Review B · Editors' Suggestion · link

2019
Band gap prediction for large organic crystal structures with machine learning
Advanced Quantum Technologies · link

Full list on Google Scholar.

Open source

550+ stars across repositories on github.com/bartolsthoorn. Highlights: NVDSP, an iOS / macOS audio DSP library (418★); NQS-numpy, neural-network quantum states in NumPy (35★); simhash-rs, a Rust implementation of SimHash (25★); gohighs, Go bindings for the HiGHS optimizer.

Skills & languages

Core	Python, Go, full-stack web (APIs & interfaces), AWS, Databricks, PostgreSQL, distributed systems, product engineering
AI / ML	Agents and agentic loops in production; PyTorch; Pandas / NumPy / scikit-learn; model training & evaluation
Past	Ruby / Rails, Elixir, Swift / Objective-C
Research	Machine learning for materials, topological data analysis, quantum systems
Languages	English, Dutch (native), Swedish (working)